# Andrew Y. Guo

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## Education

**University of Toronto** 2024 – 2029

*Doctor of Philosophy - Computer Science*

- *Advisors:* Prof. David Lindell, Prof. Kyros Kutulakos
- Focus on 3D computer vision, neural rendering, agentic AI
- Research published at CVPR 2026 as Highlight (Dark3R), Grade: 4.0/4.0

**University of Waterloo** 2019 – 2024

*Honours Bachelor of Software Engineering*

- Dean's Honour List, Cumulative Avg: 92%
- Completed 40 months of full-time work experience in industry, double the requirement for co-op program

## Publications

**Dark3R: Learning Structure from Motion in the Dark**

***Andrew Y. Guo**, Anagh Malik, SaiKiran Tedla, ..., Sotiris Nousias, Kiriakos N. Kutulakos, David B. Lindell*

CVPR 2026 (Highlight - Top 5% of Submissions) - **Paper Link** - **Project Page**

**BOP Challenge 2024 on Model-Based and Model-Free 6D Object Pose Estimation**

*Van Nguyen Nguyen, Stephen Tyree, **Andrew Y. Guo**, ..., Yann Labbe, Martin Sundermeyer, Tomas Hodan*

CVPR CV4MR Workshop 2025 (Best Paper Award) - **Paper Link** - **Project Page**

**HANDAL: A Dataset of Real-World Manipulable Object Categories with Pose Annotations, Affordances, and Reconstructions**

***Andrew Y. Guo**, Bowen Wen, Jianhe Yuan, Jonathan Tremblay, Stephen Tyree, Jeffrey Smith, Stan Birchfield*

IROS 2023 - **Paper Link** - **Project Page**

## Experience

**Research Engineering Intern** Sept. 2022 – Apr. 2024

*Nvidia* *Toronto, ON*

- Researched **3D reconstruction**, **neural scene representations**, **LiDAR novel view synthesis**, and **pose estimation**
- Created automated **6D pose** annotation pipeline for data collection in the wild, **conference paper** in **IROS 2023**
- *Advisors:* Dr. Stan Birchfield, Dr. Liila Torabi, Dr. Hammad Mazhar

**Simulation Software Engineering Intern** May 2022 – Aug. 2022

*Nvidia* *Toronto, ON*

- Designed **synthetic data generation** pipeline for 6D Pose Estimation, **beat benchmarks** in original paper
- Created custom annotator to capture semantic information from simulation to generate ground truth for training
- Sped up data generation by **150%** by asynchronously writing to **s3** bucket with multiprocessing

**Machine Learning Intern** Sept. 2021 – Dec. 2021

*Nvidia* *Santa Clara, CA*

- Created features for driver-safety system: **developer.nvidia.com/drive/drive-ix**
- Optimized 11 **convolutional neural networks** (CNNs) using **TensorRT** to run inference on limited hardware
- Transitioned team's entire **SDK** to new version of Nvidia's DRIVE OS to support integration of new camera

**Software Engineering Intern** May 2021 – Aug. 2021

*Canoo* *Torrance, CA*

- Created integrated calibration framework in **C** to update ECU calibration without recompiling main application
- **Saved** company **$30,000** in annual licensing fees by creating **Python** module to generate .a2l description files

**Autonomous Vehicle Software Intern** Jan. 2021 – Apr. 2021

*WATonomous* *Waterloo, ON*

- Created localization pipeline to match camera and **LiDAR** images with *a priori* map to navigate car without GPS
- Achieved **3x** speed up using multi-processing in C++ with **OpenMP** and lane line matching with **KD-trees**
- Utilized Carla for autonomous driving simulations when access to vehicle was not possible

**Software Engineering Intern** May 2020 – Aug. 2020
*AMD* *Toronto, ON*

- Automated analysis of memory dumps and sped up debugging by detecting anomalies and key metrics
- Created register save/restore tools that interfaced with low-level drivers for power-saving features on graphics chips

## Teaching Experience

**CSC420: Introduction to Image Understanding** Fall 2025
*Teaching Assistant* *University of Toronto*

**CSC111: Foundations of Computer Science II** Winter 2025
*Teaching Assistant* *University of Toronto*

**SE464: Software Design and Architectures** Fall 2023, Winter 2024
*Teaching Assistant and Course Curriculum Designer* *University of Waterloo*

## Awards and Scholarships

**Ontario Graduate Scholarship** 2026
- $15,000 Scholarship from the University of Toronto

**Queen Elizabeth II Graduate Scholarship in Science & Technology** 2025
- $15,000 Scholarship from the University of Toronto

**Sandford Fleming Foundation Exchange Scholarship** 2023
- $2000 Scholarship from the University of Waterloo

**American Bureau of Shipping Scholarship** 2022
- $8000 Scholarship from the University of Waterloo

**President's Research Award** 2022, 2021
- $1500 Scholarship from the University of Waterloo for Undergraduate Research
- Award twice in consecutive years

**Mensa Canada Frank & Betty Woodhams Memorial Scholarship** 2021
- $5000 Scholarship from Mensa Canada

**University of Waterloo Entrance Scholarship** 2019
- $15,000 Entrance Scholarship

**President's Scholarship of Distinction** 2019
- $2000 Entrance Scholarship from the University of Waterloo

**BC Provincial Scholarships** 2019
- Awarded $5000 in scholarships from the Government of British Columbia for undergraduate studies

**Loblaws National Scholarship** 2019
- $1500 Scholarship from Loblaws for undergraduate studies

## Volunteering

**Project Developer** May 2021 – Apr. 2022
*UW Blueprint*

- Used **React, TypeScript, GraphQL** and **MySQL** to build a translation platform for PlanetREAD, a NPO in India with **500,000 users** aiming to improve literacy by removing language barriers
- Architected database models for storing hundreds of translations, users, and stories
- **Open source** full-stack project, code can be found at: **github.com/uwblueprint/planet-read**